

Math 372: Homework 05

Due Friday, Feb 27

Instructions: Do not write your solutions on this sheet—you don't have enough space to show your work. So use separate sheets of paper.

Problem 1. The prevalence of a disease in a population is 0.15% (i.e., 0.0015). A new diagnostic test is advertised as having an false negative rate of 1% and a false positive rate of 1%. What is the probability that someone who tests positive doesn't actually have the disease?

Problem 2 (independence). Alice rolls six dice and wins if she rolls at least one six. Bob rolls twelve dice and wins if he rolls at least two sixes. Who has the greater probability of winning? *[Hint: calculate the probabilities to lose.]*

Problem 3. (You do not need to show your work for this problem.) Suppose we roll 5 six-sided dice.

(a) Compute the following probabilities

- (i) Three of the numbers are even
- (ii) At least 1 six is thrown
- (iii) No even numbers are thrown

(b) What is the expected number of sixes?

Problem 4. Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$ and $\text{Var}(X)$ for the following random variables:

(a) $X =$ (value of one roll of a 6-sided dice).

(b) A fair coin toss. $Y = \begin{cases} 1 & : \text{with probability } \frac{1}{2} \\ 0 & : \text{with probability } \frac{1}{2} \end{cases}$

(c) A biased coin toss. $Z = \begin{cases} 1 & : \text{with probability } p \\ 0 & : \text{with probability } 1 - p \end{cases}$

Problem 5. A certain brand of upright freezer is available in three different rated capacities: 16, 18, and 20 cubic feet. Let X be the rated capacity of a freezer of this brand sold at a certain store. Suppose that X has probability mass function

x	16	18	20
$p(x)$	.2	.5	.3

(a) Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, $\text{Var}(X)$.

(b) If the price of a freezer having capacity X is $70X - 650$, what is the expected price paid by the next customer to buy a freezer?

(c) What is the variance of the price paid by the next customer?

(d) Suppose that although the rated capacity of a freezer is X , the actual capacity is $h(X) = X - .008X^2$. What is the expected actual capacity of the freezer purchased by the next customer.

Problem 6 (Binomial random variable). Let $\alpha \in (0, 1)$ be a fixed parameter. Suppose we flip a coin 4 times, each time with the probability of heads being α . Let

$X =$ the number of heads out among the 4 flips.

- (a) Make a table like the following, including all 16 possible outcomes:

Outcome	Value of X	Probability
HHHH	4	α^4
HHHT	3	$\alpha^3(1 - \alpha)$
HHTH	3	$\alpha^3(1 - \alpha)$
HHTT	2	$\alpha^2(1 - \alpha)^2$
...	...	...

- (b) From now on, let's assume that $\alpha = \frac{1}{2}$. In other words, we are flipping fair coins. Recalling that $p(x) = \mathbb{P}[X = x]$, fill out a table like the following:

x	0	1	2	3	4
$p(x)$					

- (c) Verify that $\sum_x p(x) = 1$.
 (d) Compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.
 (e) Use your answer from the previous part to compute $\text{Var}(X)$.

Problem 7 (Properties of the geometric distribution). Let $\alpha \in (0, 1)$, and let X be a random variable with probability mass function

$$p(x) = \begin{cases} (1 - \alpha)^{x-1} \alpha & \text{if } x = 1, 2, 3 \dots \\ 0 & \text{otherwise.} \end{cases}$$

(A random variable with this probability mass function is called a *geometric random variable* with success parameter α .)

- (a) We know that in general probability mass functions should always sum up to one. Let's check that it is indeed the case here. Specifically, carefully show that $p(1) + p(2) + p(3) + \dots = 1$, making sure to justify your steps.
 (b) Next we will compute the expected value of X using the formula

$$\mathbb{E}[X] := \sum_x xp(x),$$

where x ranges over all possible values of X .

- (i) Expand out above summation, including the first five terms followed by “+...”
 (ii) The geometric series formula tells us that as long as $|x| < 1$, the following formula holds:

$$a + ax + ax^2 + ax^3 + ax^4 + ax^5 + \dots = \frac{a}{1 - x}.$$

Differentiate both sides of the above equation with respect to x .

- (iii) Use your answers to parts (i) and (ii) to deduce the value of $\mathbb{E}[X]$
 (iv) What is the value of $\mathbb{E}[X]$ if $\alpha = \frac{1}{2}$? What about if $\alpha = \frac{2}{3}$?